

Remove stopwords from the text file

Find top 5 frequently Occurred Words in input text file

```
rdd1 = sc.textFile("../data/Einstein.txt")
Words = rdd1.flatMap(lambda line:line.split(" "))
Words.count()
467
WordsUpper = Words.map(lambda u:u.upper())
WordsUpper.count()
467
```

```
import nltk
>>> nltk.download('stopwords')
```

move englishstopwords into data folder renamed as
english_stopwords

```
stopwords = sc.textFile("../data/english_stopwords")
>>> stopwords.count()
198
StopwordsUpper = stopwords.map(lambda st:st.upper())
>>> StopwordsUpper.count()
common = WordsUpper.intersection(StopwordsUpper)
>>> common.count()
31
TextWithoutStopwords = WordsUpper.subtract(common)
>>> TextWithoutStopwords.count()
275
TextWithoutStopwords.saveAsTextFile("../data/Results/res1.txt")
wordCounts = TextWithoutStopwords.map(lambda word: (word,1))
>>> wordCounts.count()
273
```

Basically TextWithoutStopwords and wordCounts are same
wordCounts1 = wordCounts.reduceByKey(lambda a, b: a + b)
wordCounts1.saveAsTextFile("../data/Results/wordCounts1.txt")

```
top_words = wordCounts1.sortBy(lambda x: x[1], ascending=False)
print(top_words.collect())
```

```
top5 = wordCounts1.takeOrdered(5, key=lambda x: -x[1])
>>> print(top5)
[('THEORY', 5), ('PHYSICS', 4), ('EINSTEIN', 4), ('GERMAN', 3),
('QUANTUM', 3)]
```